

Machine Learning Algorithms Activity Day

Guidelines

Objective

Each group is assigned a Machine Learning algorithm to study, understand, and present to the class. The purpose of this activity is to strengthen your theoretical understanding, improve your technical presentation skills, and demonstrate the practical implementation of your assigned algorithm.

Each group is expected to deliver:

1. A presentation explaining the algorithm.
2. A Notebook demonstrating its implementation.
3. (Optional) A short Kahoot quiz to evaluate your classmates' understanding.

Deliverables

1. Presentation (Required)

Your presentation should be well-organized, visually appealing, and technically accurate. It should cover the following topics.

1. Introduction

- What is the algorithm?
- Why was it developed?
- What problem does it solve?
- Where is it commonly used?

2. Theoretical Explanation

Explain the algorithm in sufficient detail.

Include:

- The intuition behind the algorithm
- How it works step-by-step
- Mathematical intuition (only when appropriate)
- Important terminology
- Model architecture or workflow

Use diagrams and visual illustrations whenever possible.

3. Regression and Classification

Explain whether your algorithm supports:

- Classification
- Regression
- Both

If both are supported, explain:

- How the algorithm differs between regression and classification.
- The loss/objective function used in each case.
- Output interpretation.

4. Hyperparameters

Discuss the most important hyperparameters, such as:

- learning_rate
- max_depth
- n_estimators
- min_child_weight
- subsample
- regularization parameters
- etc.

Explain:

- What each parameter controls
- How changing it affects the model
- Practical recommendations

5. Advantages

Discuss the strengths of the algorithm.

Examples:

- Accuracy
- Speed
- Interpretability

- Handling missing values
- Robustness
- Scalability
- Feature importance

6. Limitations

Discuss the disadvantages.

Examples:

- Computational cost
- Overfitting
- Large memory usage
- Sensitive hyperparameters
- Long training time
- Difficulty in interpretation

7. Best Use Cases

Explain:

- Which datasets are suitable for this algorithm?
- What type of features does it handle well?
- Small vs. large datasets
- High-dimensional data
- Imbalanced datasets
- Structured vs. unstructured data
- Sparse data
- Categorical features
- Missing values

Provide real-world applications whenever possible.

Examples:

- Medical diagnosis
- Credit scoring
- Fraud detection
- Recommendation systems
- House price prediction
- Customer churn prediction

8. Comparison with Similar Algorithms

Compare your algorithm with similar approaches.

For example:

- LightGBM vs Random Forest
- Decision Tree vs Extra Trees
- Etc...

Discuss:

- Performance
- Speed
- Memory usage
- Advantages
- Limitations

9. Feature Importance (if applicable)

Explain:

- How feature importance is computed.
- Why it is useful.
- Show a feature importance plot if applicable.

10. Practical Tips

Mention practical recommendations, such as:

- Data preprocessing requirements
- Handling missing values
- Encoding categorical variables
- Scaling requirements
- Hyperparameter tuning strategies
- Common mistakes to avoid

2. Coding Notebook (Required)

Prepare a clean and well-documented Jupyter Notebook.

The notebook should include:

Dataset Introduction

- Brief description
- Source
- Number of samples
- Features
- Target variable

Exploratory Data Analysis (EDA)

Perform basic EDA including:

- Missing values
- Data types
- Distribution plots
- Correlation analysis (if applicable)
- Target distribution

Data Preprocessing

Perform all necessary preprocessing steps, such as:

- Missing value handling
- Encoding
- Feature engineering (if needed)
- Train/Test split
- Scaling (only if required)

Explain why each preprocessing step is necessary for your algorithm.

Model Training

Train your assigned algorithm.

Include:

- Default model
- Tuned model (recommended)

Hyperparameter Tuning

Use techniques such as:

- GridSearchCV
- RandomizedSearchCV

Explain why you selected the chosen parameters.

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Model Evaluation

For Classification:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC (if applicable)
- Confusion Matrix

For Regression:

- MAE
- MSE
- RMSE
- R^2 Score

Interpret the evaluation results instead of only displaying the numbers.

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Feature Importance / Model Interpretation

If supported by your algorithm, include:

- Feature Importance
 - SHAP values (optional)
 - Partial Dependence Plots (optional)
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Conclusions

Summarize:

- What you learned
 - Model strengths
 - Model weaknesses
 - Possible improvements
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Presentation Tips

- Every member of the group should participate in the presentation.
- Avoid reading directly from the slides.
- Explain concepts in your own words.
- Use diagrams and animations when appropriate.
- Include examples to make concepts easier to understand.
- Keep your slides clean and avoid excessive text.
- Be prepared to answer questions from the instructor and your classmates.

Optional Bonus Activity

Kahoot Quiz

Groups are encouraged to create a short Kahoot quiz (5–10 questions) after their presentation.

The goal is to:

- Check whether your classmates understood the presented material.
- Make the session more interactive and engaging.

Questions should focus on:

- Key concepts
- Advantages and disadvantages
- Appropriate use cases
- Important hyperparameters
- Practical scenarios

Bonus credit may be awarded for well-designed and engaging quizzes.

Evaluation Criteria

Your work will be evaluated based on:

Criterion	Weight
Theoretical understanding	25%
Presentation quality and organization	15%

Technical accuracy	15%
Practical implementation (Notebook)	20%
Code quality and documentation	10%
Ability to answer questions	10%
Team participation	5%

Bonus:

- Interactive Kahoot quiz
- Additional visualizations
- SHAP analysis or advanced interpretation
- Demonstrating real-world applications
- Comparing results with another algorithm

Important Notes

- Understand the algorithm thoroughly instead of memorizing information.
- Use reliable references and cite your sources.
- Test your notebook before submission to ensure it runs without errors.
- Clearly explain every important decision made during preprocessing and model development.
- Focus on teaching your classmates not just presenting information.
- Creativity, clarity, and effective communication are highly encouraged.